Sequence 12

COORDINATE GEOMETRY: VECTORS AND LINES SYNTHESIS

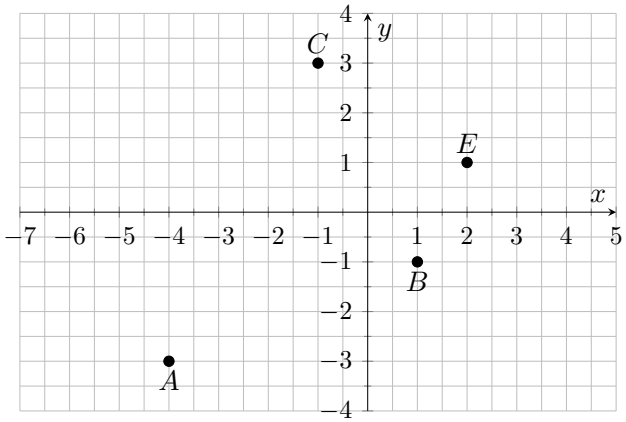
15–26 March 2027

I Parallelogram and intersection

Example 12.1: Booklet synthesis

In an orthonormal frame, $A(-4;-3)$, $B(1;-1)$ and $C(-1;3)$.

1. Determine D such that $ABCD$ is a parallelogram.
2. Determine its centre I in two ways.
3. With $E(2;1)$, determine K, the intersection of (AB) and (CE).
4. Determine F such that $\overrightarrow{AF} = \frac{2}{3}\overrightarrow{AE}$. Are D, F, B collinear?



Solution:

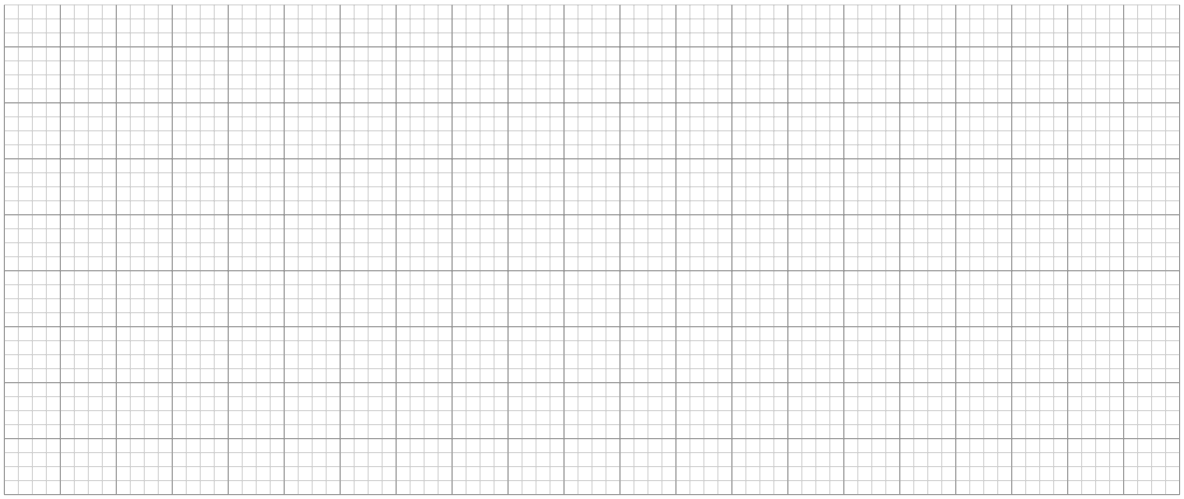
II Comparing a vector method and a coordinate method

Example 12.2

In a non-degenerate triangle ABC, I is the midpoint of $[AB]$ and J the midpoint of $[AC]$.

1. Prove with vectors that $(IJ) \parallel (BC)$.
2. Check with $A(0;0)$, $B(6;0)$, $C(2;4)$.
3. For these coordinates, determine the equations of (IJ) and (BC), then the ratio IJ/BC.

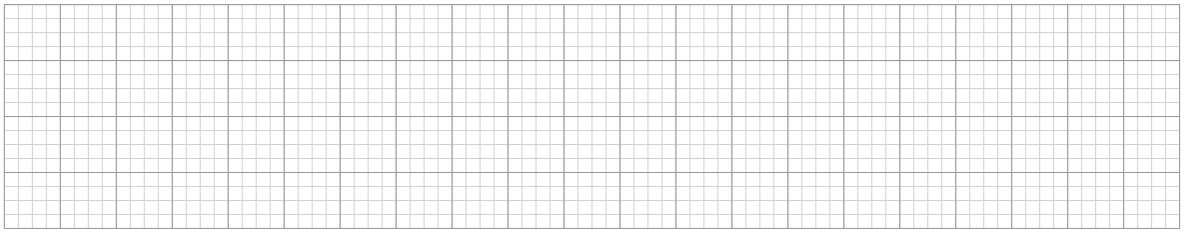
Solution:



Example 12.3

A figure seems to show two parallel lines. What is the difference between this observation and a proof?

Solution:



III Recognising a configuration and justifying

Example 12.4: Original situation

In an orthonormal frame, $A(0;0)$, $B(4;2)$, $C(3;4)$ and $D(-1;2)$.

1. Show that $ABCD$ is a parallelogram.
2. Calculate the four lengths and determine whether it is a rhombus.
3. Calculate AC^2, BD^2 and determine whether it is a rectangle.
4. Determine its centre and an equation of each diagonal.

Solution:

